

Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

Zheng Huang · Topics of Central Banking · October 2026

Speaker script · 20 minutes + 10 minutes Q&A

THE STORY, SLIDE BY SLIDE

- 2 This talk follows one question from stimulus vouchers to the design of the digital euro.
- 3 When the central bank creates money that nobody spends, monetary policy loses its grip.
- 4 Interest rates could not be cut far below zero, because cash always pays zero.
- 5 Vouchers with an expiry date get money spent, which is exactly what a liquidity trap needs.
- 6 A programmable digital euro could build this pressure into money itself, in one of two ways.
- 7 To compare the two designs, consider a holder who must search for good uses of money.
- 8 An expiry date pushes out more spending, but it ends in a fire sale and splits money into vintages.
- 9 Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate.
- 10 Yet the draft regulation fixes the digital euro's interest rate at exactly zero.
- 11 A zero-rate digital euro is cash without storage costs, so it pushes the lower bound back up to zero.
- 12 The lower bound and disintermediation are the same problem: a central-bank rival to bank deposits.
- 13 A hard holding limit stops a run in a crisis, but it also blocks ordinary users in normal times.
- 14 When costs have a cliff and benefits do not, price and quantity together beat either one alone.
- 15 Each segment of the cost curve becomes one rule: a free tier, a Pigouvian rate, and a cap.
- 16 In an illustrative calibration, the rules imply a free tier of about €500, a rate of about -2.4% and a cap near €3,000.
- 17 If money is to carry a holding cost, continuous decay with a free tier and a cap is its natural form.

TIMING

1	0:15	10	0:40
2	0:45	11	1:10
3	1:15	12	1:00
4	1:15	13	1:15
5	1:15	14	1:20
6	0:50	15	1:30
7	1:15	16	1:20
8	1:30	17	1:10
9	1:15		

Checkpoints: end of slide 6 at 5:35; slide 9 at 9:35; slide 13 at 13:40; slide 17 at 19:00, leaving one minute of slack.

NUMBERS TO KNOW

- Eurosystem balance sheet: €2.2tn (2014) to €8.6tn (2021); inflation below 2% in 2014–20.
- Deposit facility rate ≤ 0 from June 2014 to July 2022; lowest -0.50% (2019–22).
- China coupons: ¥3 per ¥1. France: one-month MPC 61% vs 23%.
- Expiry vs decay, 30 days: 84% vs 73% useful spending; 16% dumped; value 0.80 to 0.50.
- Decay of 1% a year raises velocity by about 12%.
- Art. 16(8): “shall not bear interest”. Cap range €1,500–3,000.
- Crisis outflow: €699bn at €3,000 (ECB); model €648bn; no cap €4.7tn.
- Illustration: free tier €510, decay -2.4% ($2.65\% - 0.26\%$), cap €3,000.
- Appendix: design comparison (cap only 18.5% blocked; price only €4.7tn in a crisis).

SLIDE 1

0:15

Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

Zheng Huang
Topics of Central Banking
October 2026

SAY

- “Good morning. My talk is about what happens if a central bank programs a holding cost into money, and what that means for the design of the digital euro.”

SLIDE 2

0:45

OVERVIEW

This talk follows one question from stimulus vouchers to the design of the digital euro

Should the digital euro bear interest, and if so, in what form?

- THE PROBLEM
Monetary policy stalls when new money is hoarded and interest rates cannot fall below the cash floor.
- A TEMPTING FIX
Vouchers with an expiry date get money spent, and a programmable digital euro could carry one.
- A BETTER FIX
An expiry date ends in a fire sale and splits money into vintages; continuous decay is simply a negative rate.
- THE TWIST
The draft regulation fixes the rate at zero, which makes the digital euro costless cash and raises the lower bound.
- THE DESIGN
Combining price and quantity suggests a free tier for payments, a Pigouvian rate above it, and a cap for crises.

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SAY

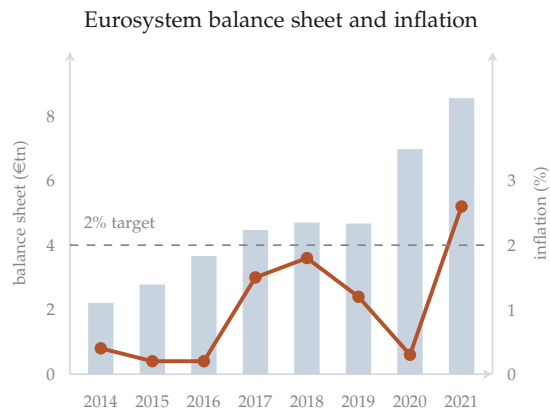
- “The whole talk answers one question: should the digital euro bear interest, and in what form?”
- Walk down the five steps in one sentence each; tell the audience the answer comes in three numbers at the end.

SLIDE 3

1:15

THE PROBLEM · 1

When the central bank creates money that nobody spends, monetary policy loses its grip



Sources: ECB annual consolidated balance sheet of the Eurosystem (end of year); Eurostat HICP, annual average.

Between 2014 and 2021 the Eurosystem almost quadrupled its balance sheet, from €2.2 trillion to €8.6 trillion.

Inflation nevertheless stayed below the 2% target in every year from 2014 to 2020, and was close to zero in 2014–16 and 2020.

In the quantity equation $MV = PY$, the extra money M was offset by a fall in velocity V : households and firms held the new money instead of spending it.

This is the first blockage: money is created, but it is not spent.

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SAY

- The bars are the Eurosystem balance sheet: almost four times larger in seven years.
- The line is inflation: below target every year until 2021.
- In $MV = PY$, more M did not raise PY because V fell. People held the money.
- “That is the first blockage: money is created but not spent.”

IF ASKED

Q Wasn't the balance sheet growth meant to lower long-term rates, not to raise spending directly?

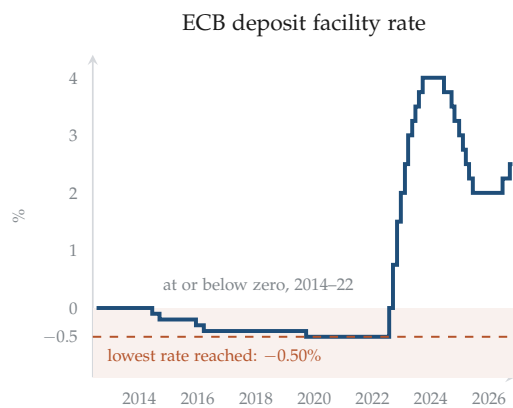
A Yes, asset purchases worked through yields and portfolio rebalancing. The point is only that the extra base money did not translate into proportional spending.

SLIDE 4

1:15

THE PROBLEM · 2

Interest rates could not be cut far below zero, because cash always pays zero



Source: ECB key interest rates.

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The ECB cut its deposit facility rate below zero in June 2014 and kept it at or below zero until July 2022, but it never went lower than -0.50% .

The reason is the outside option of cash. Banknotes pay zero, so if banks charged savers more than the cost of storing and insuring cash, savers would withdraw their money.

The lower bound is therefore not exactly zero but minus the cost of holding cash: $i \geq -c$.

This is the second blockage: interest rates cannot fall far enough to make people spend.

SAY

- The deposit facility rate went below zero in June 2014 and stayed at or below zero for eight years.
- It stopped at -0.50% . Below that, savers would pull money into banknotes, which pay zero.
- The true floor is minus the cost of holding cash: storage, insurance, transport.
- “That is the second blockage: rates cannot fall far enough.”

IF ASKED

Q Inflation is now above target. Why does this matter?

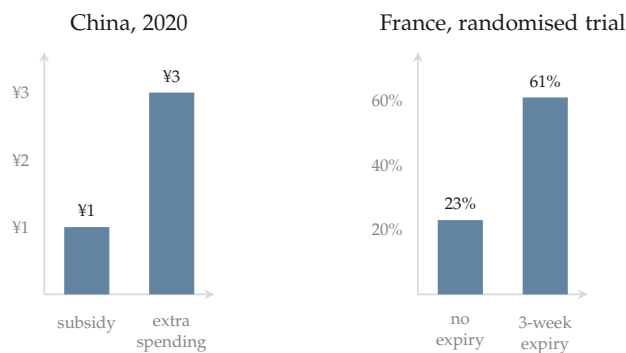
A The floor was binding for most of 2009–2021, and the regulation is likely to last decades, so its design matters in both regimes.

SLIDE 5

1:15

A TEMPTING FIX

Vouchers with an expiry date get money spent, which is exactly what a liquidity trap needs



In 2020 Chinese cities distributed digital coupons through mobile payment platforms that had to be used within about a week. Each yuan of subsidy generated roughly three yuan of additional spending.

In a randomised experiment in France, households received prepaid cards. When the balance expired after three weeks, 61% was spent within a month, against 23% without an expiry date.

A deadline turns money that would have been saved into spending today.

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024), one-month MPC.

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SAY

- Governments found a way around the first blockage: vouchers with a deadline.
- Left: Chinese digital coupons, roughly three yuan of extra spending for each yuan of subsidy.
- Right: a French randomised trial; the expiry date nearly triples the one-month spending rate.

IF ASKED

Q The Chinese coupons also had minimum-spend thresholds. Isn't that what drives the effect?

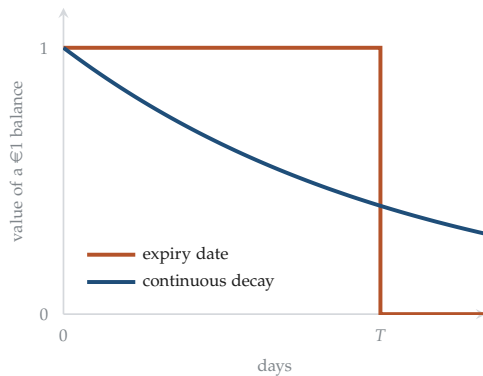
A Partly, as Ding et al. show. That is why I add the French trial, which isolates the deadline.

SLIDE 6

0:50

A TEMPTING FIX

A programmable digital euro could build this pressure into money itself, in one of two ways



A central bank digital currency is programmable, so the central bank could charge holders for keeping money idle.

With an **expiry date**, nothing happens until day T , and then the whole balance is lost.

With **continuous decay**, also called demurrage, the balance shrinks by a small fraction every day, like Silvio Gesell's stamped money of the 1930s.

Both designs charge for holding money; they differ only in *when* the charge falls. **Which one should a central bank choose?**

SAY

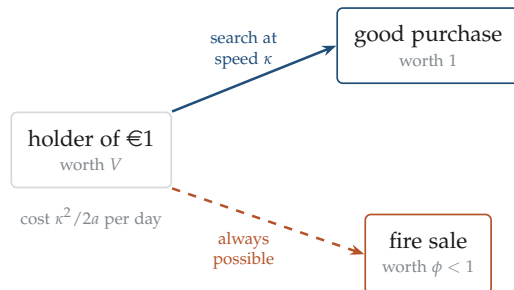
- A digital euro is programmable, so a holding cost could be written into money itself.
- Red: an expiry date; the balance is untouched until T , then gone. Blue: continuous decay, Gesell's idea; a little every day.
- "Same total cost, different timing. Which one should a central bank choose?"

SLIDE 7

1:15

THE MODEL

To compare the two designs, consider a holder who must search for good uses of money



Each euro is worth V to its holder. Finding a good use for it takes effort: searching at speed κ costs $\kappa^2/2a$ per day, and κ is the velocity of money. A bad use, such as buying something of little value or selling the token at a discount, is always available and worth $\phi < 1$.

Decay at rate λ : $(r + \lambda)V = \frac{a}{2}(1 - V)^2$

Expiry at T : $\frac{dV}{d\tau} = -rV + \frac{a}{2}(1 - V)^2$, $V = \phi$ at T

In both cases the holder spends at speed $\kappa^* = a(1 - V)$. The less money is worth holding, the faster it moves.

SAY

- To answer that, I need a simple model of how holders spend.
- A holder searches for a good use of each euro; faster search costs more. A fire sale is always possible but worth less.
- The value of holding money satisfies one equation under decay and a differential equation under expiry.
- "In both cases, the less money is worth holding, the faster it is spent."

IF ASKED

Q Why a quadratic search cost?

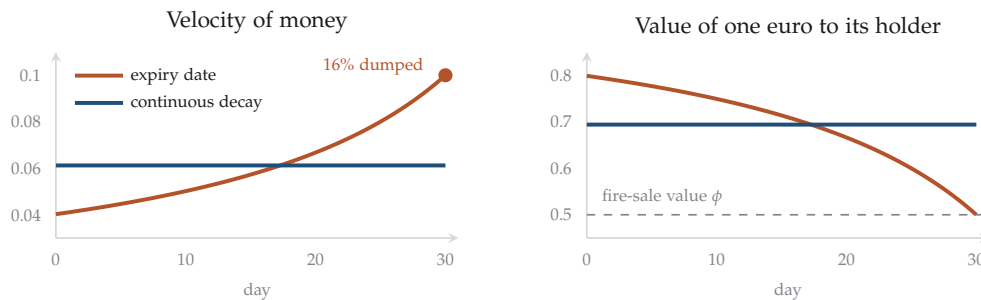
A It is the simplest convex cost with an interior solution. With a linear cost, velocity would literally explode at the deadline; convex is the conservative case.

SLIDE 8

1:40

RESULT 1

An expiry date pushes out more spending, but it ends in a fire sale and splits money into vintages



Both designs are calibrated to the same average velocity over 30 days. The expiry date gets more useful spending done, 84% of issuance against 73%, because it forces the whole balance out. But velocity accelerates into the deadline, and 16% of issuance is still unspent on the last day and is dumped on goods worth half its value. Worse, an expiring euro is worth 0.80 when new and 0.50 just before it expires: one euro is no longer equal to another.

Parameters: $a = 0.2$, $\phi = 0.5$, $r = 4\%$ a year. Results at other horizons and parameters in the backup.

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SAY

- Both designs are tuned to the same average velocity over 30 days, so the race is fair.
- Left: velocity under expiry accelerates into the deadline; under decay it is flat.
- Expiry gets more useful spending done, 84% against 73%, because it forces the whole balance out. That surprised me.
- But 16% is still unspent on the last day and goes to a fire sale.
- Right: an expiring euro is worth 0.80 when new and 0.50 when old. One euro is no longer equal to another.

IF ASKED

Q Why match average velocity rather than total spending?

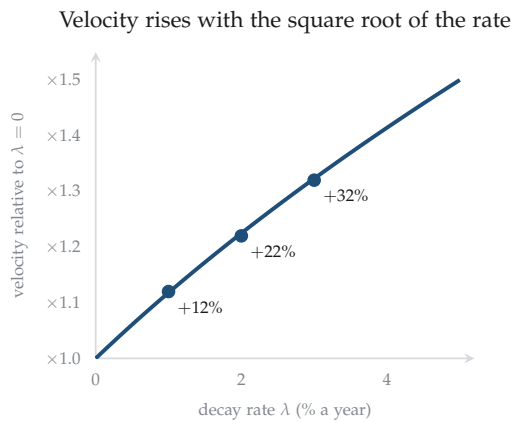
A Velocity is what the central bank wants to raise in a liquidity trap. Matching total spending is impossible, because decay always takes part of the balance.

SLIDE 9

1:15

RESULT 2

Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate



Decay avoids every problem of the expiry date. There is no deadline and no fire sale, every euro has the same value whatever its age, and it can apply to all money rather than to a separate voucher.

A decay rate λ is simply an interest rate of $-\lambda$.

Velocity follows $\kappa^* \approx \sqrt{2a(r + \lambda)}$, the same square-root rule as in the Baumol–Tobin model of money demand: a decay of 1% a year raises velocity by about 12%.

Decay addresses both blockages: money moves faster, and a negative rate finally reaches retail balances. An expiry date belongs to a voucher; decay can belong to money.

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SAY

- Decay avoids all three problems: no deadline, no fire sale, every euro worth the same.
- And decay at λ is just an interest rate of $-\lambda$.
- The curve: velocity rises with the square root of the rate, as in Baumol–Tobin. One percent of decay, 12% more velocity.
- “Decay clears both blockages: money moves faster, and a negative rate reaches retail balances.”

IF ASKED

Q Couldn't a smart contract simply track the age of each expiring euro?

A Yes, but then shops must accept old and new euros either at par, which invites arbitrage, or at different prices, which ends the singleness of money.

SLIDE 10

0:40

THE TWIST

Yet the draft regulation fixes the digital euro's interest rate at exactly zero

"The digital euro shall not bear interest."

European Commission, proposal for a regulation on the digital euro, Article 16(8), June 2023



The rate is not positive and not negative, in any circumstances. Holdings are also capped per person, with a range of €1,500–3,000 under discussion. The final text is being negotiated now, so the design question is still open.

SAY

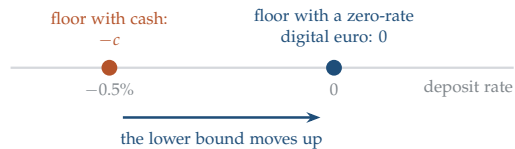
- “So the natural instrument is a rate on the digital euro. But the draft regulation rules it out.”
- Read the quote; point to the timeline. The text is in trilogue now, so the question is open.

SLIDE 11

1:10

THE TWIST

A zero-rate digital euro is cash without storage costs, so it pushes the lower bound back up to zero



A bank cannot pay depositors less than their best alternative in central-bank money, net of the cost of holding it:

$$r_D \geq \max_j \{ i_j - c_j \}.$$

Cash pays zero but costs something to store and insure, so the floor lies slightly below zero. This is why the ECB could reach -0.50% .

A zero-rate digital euro pays the same zero but costs nothing to hold. Up to the holding limit, it lifts the floor for deposit rates back to exactly zero, and takes away the little room below zero that the ECB had.

SAY

- A bank cannot pay less than the best central-bank alternative, net of the cost of holding it.
- Cash pays zero but is costly to hold, so the floor sits at $-c$. That is why -0.50% was possible.
- A zero-rate digital euro pays zero and costs nothing. Up to the cap, the floor moves back to zero.

IF ASKED

Q Retail deposit rates never went negative anyway.

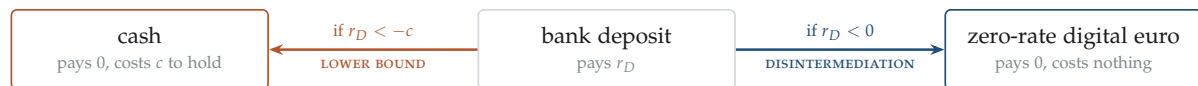
A For small balances banks chose not to charge. Large and corporate deposits did pay negative rates. A zero-rate digital euro turns zero into a legal floor for everyone, up to the cap.

SLIDE 12

1:00

THE TWIST

The lower bound and disintermediation are the same problem: a central-bank rival to bank deposits



The lower bound and bank disintermediation are usually discussed as separate issues. They are the same mechanism: whenever central-bank money offers a better net return than a bank deposit, depositors move.

A zero-rate digital euro therefore puts the most pressure on bank deposits precisely when rates are negative, which is when banks' margins are already squeezed. Fixing its rate at zero ties the digital euro's financial-stability risk to the state of monetary policy.

SAY

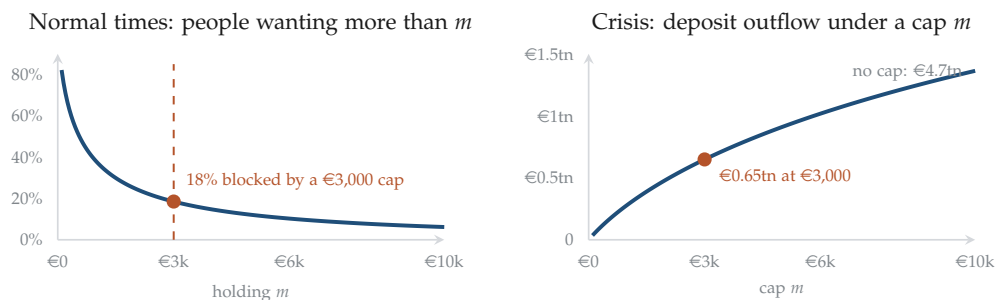
- Left arrow: deposits flow into cash when $r_D < -c$. That is the lower bound.
- Right arrow: deposits flow into the digital euro when $r_D < 0$. That is disintermediation.
- “One mechanism, two names. And with a zero rate, the risk peaks exactly when rates are negative.”

SLIDE 13

1:15

THE TRADE-OFF

A hard holding limit stops a run in a crisis, but it also blocks ordinary users in normal times



In normal times people hold the digital euro to make payments. In the ECB's survey, 18% would like to hold more than €3,000; a hard cap simply blocks them, while a price would let them hold more if they pay for it. In a crisis people run for safety, and a small interest-rate difference cannot stop them. **Price is enough in normal times; only quantity works in a crisis.**

Survey distribution: Lambert et al. (ECB WP 2980, 2024). Outflows: own calibration to ECB technical data on holding limits (Oct 2025), which reports €699bn at €3,000.

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SAY

- The ECB's current answer to disintermediation is a hard cap, around €3,000.
- Left: in normal times, about 18% of people would like to hold more; the cap simply blocks them.
- Right: in a crisis the cap is what matters: about €0.65 trillion of outflow at €3,000, against €4.7 trillion with no cap.
- "So a price seems sufficient in normal times, while in a crisis only a quantity limit works."

IF ASKED

Q Where does the crisis curve come from?

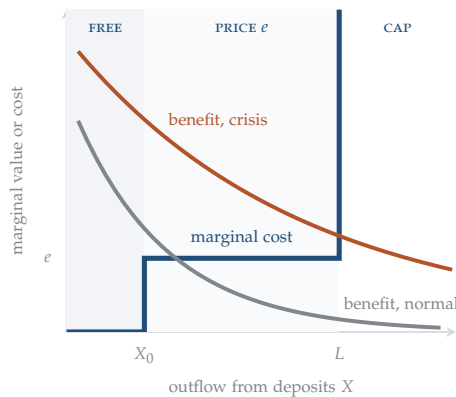
A The ECB published €156bn at a €500 cap and €699bn at €3,000. I fit the survey distribution of deposits to those two points. The implied population is 360 million, close to the actual euro-area population, and the no-cap outflow of €4.7tn is close to Meller and Soons' €5tn.

SLIDE 14

1:20

THE PRINCIPLE

When costs have a cliff and benefits do not, price and quantity together beat either one alone



Weitzman (1974) showed that under uncertainty a price works better when the benefit curve is steeper than the cost curve, and a quantity works better when the cost curve is steeper:

$$\Delta_{\text{price-quantity}} = \frac{\sigma^2 (b - c)}{2b^2}.$$

The marginal cost of deposit outflows has three segments: zero while banks can absorb them, a constant e while banks replace deposits with dearer funding, and vertical once their liquidity buffers run out.

In normal times demand crosses the flat part, where a price works best. In a crisis it shifts out and hits the cliff, where only a quantity works. **Both happen on the same curve, which points to a tiered price with a cap** (Roberts and Spence 1976).

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SAY

- This is an old question: prices or quantities? Weitzman's answer depends on slopes: a price when benefits are steeper than costs, a quantity when costs are steeper.
- The blue staircase is the cost of deposit outflows: free, then a constant e , then a cliff.
- In normal times demand crosses the flat part, where a price works; in a crisis it hits the cliff, where only a cap works.
- "Both cases sit on one curve, which points to combining the two, as Roberts and Spence showed."

IF ASKED

Q Hasn't Bindseil already proposed tiering?

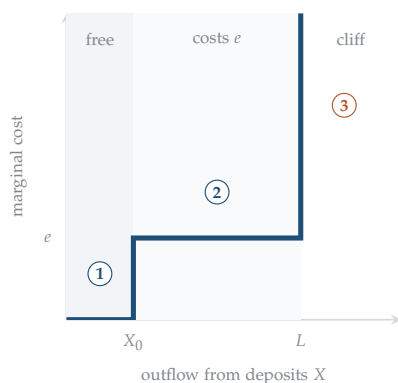
A Yes, and this builds on it. What the framework adds is a reason for each tier, from the shape of the cost curve.

SLIDE 15

1:30

THE DESIGN · THEORY

Each segment of the cost curve becomes one rule: a free tier, a Pigouvian rate, and a cap



1. A free tier M_1 . Outflows up to X_0 cost banks nothing, so balances held for payments can be free. The tier is set so that payment balances just fill this free capacity: $NE[\min(t, M_1)] = X_0$, where t is what a person wants to hold.

2. A Pigouvian rate $-e$ above the tier. Each further euro moved out of a bank costs the bank e to replace. Charging holders exactly this cost makes them weigh it, as a Pigouvian tax does: $i = -e$.

3. A cap $\bar{M}$. Beyond the liquidity buffer L the cost has no bound, so no price is high enough. Only a quantity limit keeps the crisis outflow within the buffer: $NE[\min(d, \bar{M})] = L$, where d is a person's sight deposit, all of which may run.

SAY

- Before any numbers, the logic: each segment of the staircase turns into one rule.
- Segment 1, free: outflows that banks absorb at no cost. So payment balances can be free, and the tier is set so that they just fill that free capacity.
- Segment 2, cost e : each further euro costs the bank e to replace. A Pigouvian rate of $-e$ makes holders bear that cost.
- Segment 3, the cliff: no price is high enough, so only a cap keeps crisis outflows within the buffer.

IF ASKED

Q What does Pigouvian mean here?

A A charge equal to the cost one person's choice imposes on others. Here the cost falls on banks, which must replace lost deposits with dearer funding.

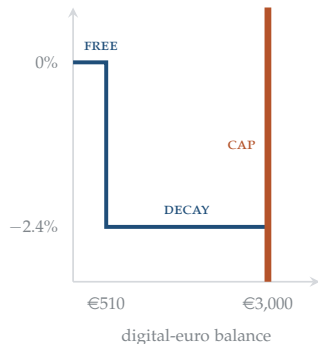
SLIDE 16

1:20

THE DESIGN · ILLUSTRATION

In an illustrative calibration, the rules imply a free tier of about €500, a rate of about -2.4% and a cap near €3,000

The resulting rate schedule



Rule	What the data say	Result
1. Free tier	The ECB expects payment digitalisation to bring €127bn of new deposits by 2034, so $X_0 = €127\text{bn}$. Desired holdings t : median €560, mean €3,220.	€510
2. Rate	A bank replaces a lost deposit at the refinancing rate (2.65%) instead of the deposit rate (0.26%), so e is 2.4 points.	-2.4%
3. Cap	Sight deposits d : median €2,000, mean €13,100. L is the crisis outflow the ECB judged manageable in October 2025.	€3,000

The cap is anchored to the ECB's own analysis. The free tier and the rate follow from the model and depend on its assumptions; **they show orders of magnitude, not a recommendation**. Details and sensitivity are in the appendix.

Data: Lambert et al. (ECB WP 2980, 2024), survey distributions; ECB technical data on holding limits (Oct 2025); ECB key rates (Sep 2026); MIR statistics (Apr 2026).

Population $N = 360\text{m}$, fitted to the ECB's outflow estimates.

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SAY

- Now the numbers, as an illustration of orders of magnitude.
- Free tier: the ECB expects digitalisation to bring €127bn of new deposits. Filling exactly that with payment balances gives a tier of about €500.
- Rate: a bank replaces a lost deposit at the refinancing rate, 2.65%, instead of 0.26%; that difference is about 2.4 points.
- Cap: I take the outflow the ECB itself judged manageable, which brings us back to its €3,000.
- “The cap reproduces the ECB's figure; the tier and the rate depend on the model's assumptions, and the appendix shows how sensitive they are.”

IF ASKED

Q Isn't -2.4% a very large penalty?

A Almost nobody would pay it; it mainly keeps the digital euro from becoming a savings account. It is also an upper bound: if central-bank funding replaces deposits cheaply, the social cost is lower.

Q Why €127bn?

A It is a natural benchmark: banks end up no worse off than today. With €100–300bn the tier ranges from €370 to €1,840 (appendix).

Q Does the hybrid actually perform better?

A In the model, yes: compared with a cap alone it blocks fewer users and leaks far less when rates are negative, and unlike a price alone it holds in a crisis. The comparison is in the appendix.

SLIDE 17

1:10

CONCLUSION

If money is to carry a holding cost, continuous decay with a free tier and a cap is its natural form

1. Deadlines get money spent, but in the model an expiry date ends in a fire sale and makes one euro worth more than another. It suits a voucher better than money.
2. Continuous decay keeps every euro equal and is equivalent to a negative interest rate, so it is the natural form for a holding cost on money.
3. A digital euro fixed at zero behaves like cash without storage costs. It tends to raise the lower bound, and its pressure on bank deposits is greatest when rates are negative.
4. Weitzman's logic suggests combining price and quantity: a free tier for payments, a rate above it, and a cap for crises.

OPEN QUESTIONS

General-equilibrium effects on prices and output; the social rather than private cost of deposit outflows; behavioural responses to a shrinking balance; and the legal and political feasibility of a negative rate on public money.

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SAY

- Summarise the four points in one breath each.
- Mention the open questions honestly: general equilibrium, the true social cost of outflows, behaviour, and legal feasibility.
- Close: "The zero-rate design is a choice with a cost. My aim was to make that cost visible, and to show what form a holding cost on money would naturally take. Thank you."

IF ASKED

Q *Should the cap move over time?*

A Possibly. Since the cap protects bank liquidity, it could follow a published rule tied to banks' liquidity ratios. I sketch this in the appendix, but it raises its own questions about signalling in a crisis.

MORE QUESTIONS

Q *Isn't negative interest on CBDC already in Bordo and Levin (2017)?*

A Yes, the idea is theirs. This talk adds a comparison of expiry and decay in one model, and a framework for combining a rate with a cap; the related literature is in the appendix.

Q *Would people accept a digital euro that can shrink?*

A Balances up to the free tier never shrink. Above it, the charge only makes the digital euro no better than a bank account.

Q *The ECB says the digital euro will not be programmable money.*

A That means no restrictions on what it can buy. Decay restricts nothing; it is a rate on a balance.

Q *Your model is partial equilibrium.*

A Yes. The qualitative results do not depend on general equilibrium, and the magnitudes are calibrated to the ECB's own numbers.

Q *Do coupons just pull spending forward?*

A The evidence is mixed: Xing et al. (2023) find purchases moved forward, Ding et al. (2024) do not. The argument does not rely on it.

GLOSSARY

Velocity	Spending per unit of money per period, $V = PY/M$.
MPC	Share of an extra euro spent within a period.
Lower bound (ZLB / ELB)	The rate below which savers switch to cash; slightly below zero because cash is costly to hold.
Demurrage / decay	A continuous charge on holding money; a negative interest rate.
Singleness of money	Every form of a currency exchanges one-for-one at par.
Baumol–Tobin	Inventory model of money demand; holdings fall with the square root of the holding cost.
Disintermediation	Deposits leaving banks for central-bank money.
Weitzman (1974)	Prices beat quantities when benefits are steeper than costs, and vice versa.
Roberts–Spence (1976)	A price schedule with a quantity ceiling beats either instrument alone.
Pigouvian rate	A charge equal to the cost one holder imposes on others.
LCR	Liquidity coverage ratio: liquid assets over 30-day stressed outflows; minimum 100%.
MRO	Main refinancing operations rate, 2.65% since 16 Sep 2026.
Trilogue	Negotiation between Commission, Council and Parliament on EU legislation.
Art. 16(8)	Draft regulation: “The digital euro shall not bear interest.”